

Millikan Oil Drops: Experiment II

Connor Fricke (fricke.59@osu.edu)

Physics 5700

17 Feb 2025

Abstract—For hundreds of years, the electron’s existence has been well known, or at least formally theorized. In this experiment, we exploit the methods employed by R.A. Millikan in the early 1900s in order to similarly demonstrate the quantization of electrostatic charge. We obtain a quantitative estimate of the charge on individual electrons within a small range of the modern accepted value, e , by recording the static charge present on atomized oil droplets between capacitor plates. Our experimentally determined results imply an electron charge of $e = (1.58 \pm 0.06) \times 10^{-19} \text{C}$.

I. INTRODUCTION

Prior to discovery of the electron, electricity was thought to be a sort of fluid. For many years, electricity was an elusive phenomenon that was not well understood. Today, we explicitly manipulate electrons at a quantum scale to create previously unimaginable technologies like quantum computers and electron microscopes. The original oil drop experiments conducted by R.A. Millikan (see Ref. [1]) between 1909 and 1913 were seminal to this development over the last hundred years or so.

By the 1800’s, scientists were already beginning to conceive of the electron’s existence. Many thought there must be an “atom of electricity”, which quantized the interactions facilitated by the electrical force. The English physicist JJ Thompson is credited with the actual discovery of the electron (1897) and an estimate of the charge to mass ratio, $\frac{e}{m}$. However; Millikan, inspired by previous attempts to measure the electron charge, was able to conduct his famous oil drop experiment in which he demonstrated this quantization and obtained an estimate of the electron’s charge, e (see Ref. [2]). Millikan’s findings put an estimate on the charge to be

$$e = 4.891 \times 10^{-10} \quad (1)$$

which was roughly equal to $e = 1.5924 \times 10^{-19}$ Coulombs, missing the currently accepted value by 0.6% (see Ref. [1]).

In replicating this experiment, it will be our goal to find similar results to the original Millikan findings and the accepted current day value of

$$e = 1.602176634 \times 10^{-19} \text{ C} \quad (2)$$

by measuring the terminal velocity of oil drops in the presence of viscous drag, gravity, and electrostatic forces.

II. THEORY

The theory behind this experiment is fairly straightforward, and is adapted here from Ref. [2]. Atomized oil droplets are sprayed between the plates of a high voltage capacitor, gaining an electric charge, q , due to friction with the air.

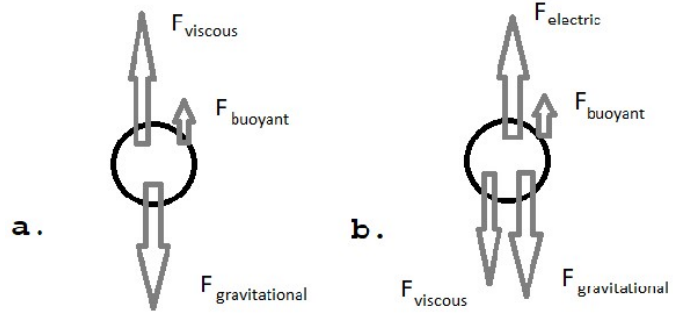


Fig. 1: Free-body diagram visualization of forces on oil droplets with (b) and without (a) the presence of an electric field. The buoyancy force is small enough to be ignored in both cases. The viscous drag force opposes the direction of motion. Figure adapted from Ref. [2].

With the capacitor discharged ($V = 0$), the droplets only experience gravity, viscous drag, and negligible buoyancy forces. A high voltage on the capacitor creates an electric field within the region of interest which allows for the experimenter to manipulate the free fall velocity of the oil droplets. Thus, there are four forces of concern:

$$F_g = mg = \rho V_d g \quad (\text{gravity}) \quad (3)$$

$$F_v = 6\pi\eta av \quad (\text{viscous drag}) \quad (4)$$

$$F_B = \rho_{\text{air}} V_d g \quad (\text{buoyancy}) \quad (5)$$

$$F_E = qE \quad (\text{electrical force}) \quad (6)$$

where ρ is the density of the oil, V_d is the volume of the drop (which is not identical for all drops), η is the viscosity of air (a function of temperature), a is the drop radius, v is the drop velocity, q is the total electrical charge on the drop, and $E = V/d$ is the magnitude of the electric field between the plate capacitor at voltage V and plate separation d .

The buoyancy force is negligible because the density of the oil is much greater than the density of air ($\rho \gg \rho_{\text{air}}$, $F_g \gg F_B$).

Without an electric field, setting equations (3) and (4) equal and assuming terminal velocity, v_g , gives

$$mg = 6\pi\eta av_g \quad (7)$$

which allows a determination of the drop radius,

$$a = \left(\frac{9\eta v_g}{2g\rho} \right)^{1/2} \quad (8)$$

since drop radius is too small to be measured directly.

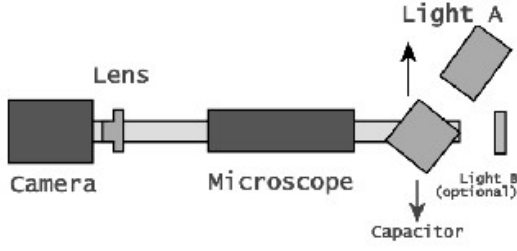


Fig. 2: Basic schematic of observation equipment setup, adapted from Ref. [2]. Atomizer and oil droplet delivery not depicted (see Fig. 3).

Without changing our assumptions, and accounting for the presence of an electric field, we balance the forces such that

$$qE = mg + 6\pi\eta av_E \quad (9)$$

where now we have v_E , which is the new terminal velocity of the drop. Equations (7) and (9) combine for a value of the charge on the drop based entirely on measurable quantities,

$$q = \frac{6\pi d}{V} \sqrt{\frac{9\eta^3}{2\rho g}} (v_E + v_g) \sqrt{v_g} \quad (10)$$

Importantly, we must correct the drag force (Stokes' Law) in the limit of extremely small spheres, which adds a factor of $(1 + \frac{b}{aP})^{-1}$ where P is the atmospheric pressure and b is an empirical constant.

$$q = \frac{6\pi d}{V} \sqrt{\frac{9\eta^3}{2\rho g}} (v_E + v_g) \sqrt{v_g} \left(1 + \frac{b}{aP}\right)^{-1} \quad (11)$$

where $b = 6.17 \times 10^{-6} \text{ m} \cdot (\text{cmHg})$.

III. METHODS

The original experiment by Millikan was inspired by previous attempts to measure the charge of an electron using water droplets, though Millikan made a significant improvement to the experiment by using oil instead of water. The reason for the change being that water droplets would evaporate over the duration of the experiment. Evaporation made impossible the observation of a single drop for more than a minute, thus oil made a much better droplet candidate for precise observations.

A camera, lens, and microscope were aligned parallel to each other on an optical rail such that the oil droplets can be observed easily within the region of interest (between the plates of a capacitor charged by a high voltage power supply, with a switch for quickly setting the polarity). An overhead view of the alignment is shown in Figure 2, as well as a depiction of the original experiment setup in Figure 3.

The capacitor itself is on a translation stage which allows for rotation of the plates as well as translation perpendicular to the optical rail, which facilitates the alignment process. The ideal angle of the capacitor is between 30 and 35 degrees relative to the optical rail, and the plates should be level.

In the center of the top plate of the capacitor is a vent which allows for droplets of oil to enter the observation chamber between the plates. The atomizer was used to spray droplets

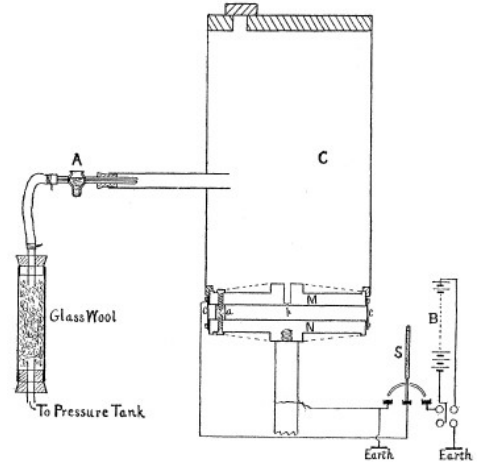


Fig. 3: Original Millikan Oil drop experimental setup. Atomizer (A) is shown to the left of the dust-free chamber (C) with the plate capacitor shown by M and N. Our experiment will use a similar design, but the atomizer is a handheld device with a small oil reservoir attached.

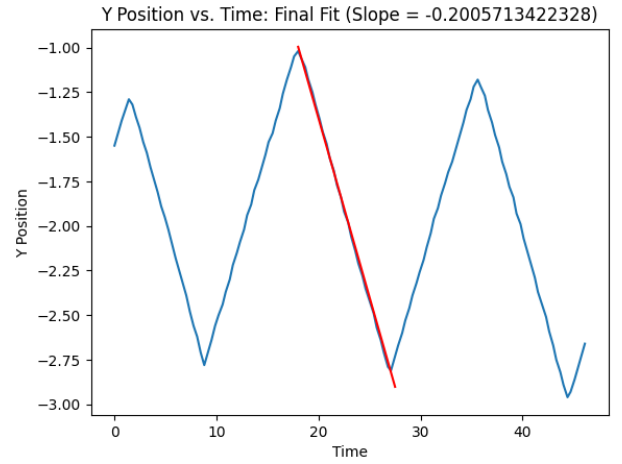


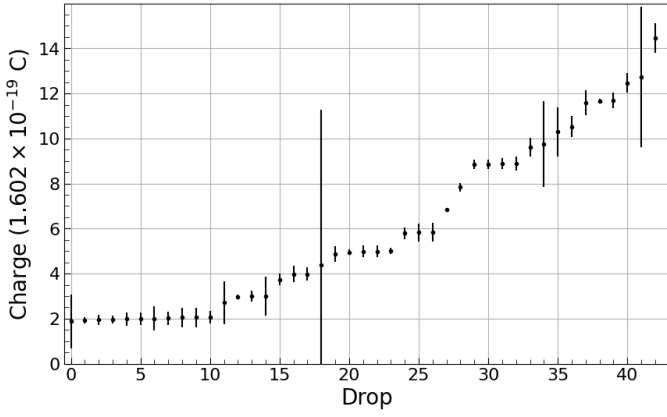
Fig. 4: Example fit for a region of data in which the drop movement is influenced by the applied electric field. Positive slopes (increasing y position) indicate the capacitor is discharged and the particle is experiencing free fall (the view through the microscope is inverted from the expectation).

of oil above the capacitor such that they are in free fall when they enter the chamber, and can easily be observed with the camera.

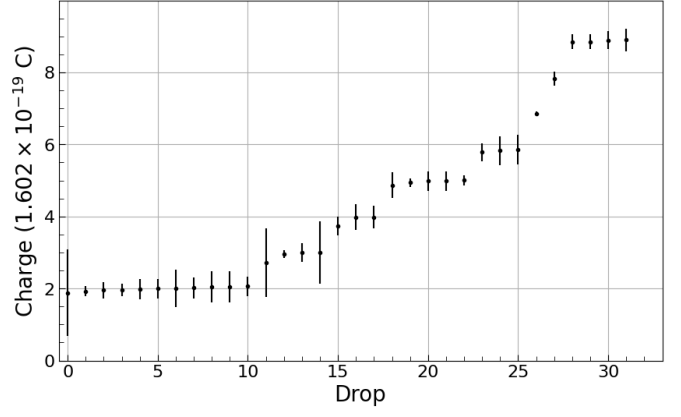
After calibrating the units of the length scale, a few dozen videos of drops moving through the chamber were recorded. In post, Tracker software was used to plot the position of the oil droplets against time to calculate both v_E and v_g through several cycles of free fall and applied electric field. An accurate estimate of drop radius is obtained with equation (8).

IV. ANALYSIS & INTERPRETATION

By fitting a slope to several cycles of position versus time data, accurate estimates for v_E and v_g are straightforward to obtain. An example of this data is shown in Figure 4.



(a) Drop charge plotted for all drops with $q < 15e$. The vertical axis is shown in units of the accepted value of the electron charge, $e = 1.602 \times 10^{-19}$ C. As expected, we see a step plot, where drops are grouped near integer multiples of e .



(b) New dataset with outlier points removed, as well as those with a large number of electrons. Once again, we see the same grouping of points near integer multiples of the electron charge. We expect that this new dataset will produce a smaller uncertainty on e .

Fig. 5

quantity	value	units	description
d	4.44 ± 0.02	mm	capacitor separation
V	400.5 ± 0.8	Volts	capacitor voltage
η	18.13×10^{-6}	$\text{N} \cdot \text{s}/\text{m}^2$	air viscosity
ρ	0.86×10^{-3}	kg/m^3	watch oil density
g	9.801	m/s^2	accel. of gravity
b	$(6.10 \pm 0.01) \times 10^{-6}$	$\text{m} \cdot \text{cmHg}$	Stokes' law corr.
P	76.0	cmHg	atm. pressure

TABLE I: Given and obtained values of constants in the droplet charge equation.

The regions in which the drop moves “upward” are actually the regions in which the drop experiences free fall, with no applied electric field. When the drop switches directions and instead begins moving “downward”, then we understand that the capacitor must be turned on, creating an electric field such that the negatively charged droplets move against gravity.

Repeated measurements of the voltage applied to the capacitor during operation produce no statistical variation, but systematic inaccuracies of the multimeter used to display the voltage yields [3]

$$V = -400.5 \pm 0.8 \text{ Volts} \quad (12)$$

This value is a constant which can be used directly in equation (11), with standard error propagation techniques. The distance between capacitor plates was measured with calipers, roughly

$$d = 4.44 \pm 0.02 \text{ mm} \quad (13)$$

Except for voltage, V and capacitor separation, d , the values in Table I are given constants. Three of these values (d, V, b) have error, meaning that there are five total sources of error for propagation (v_E and v_g have uncertainty as well). It's important to minimize error such that the error on each droplet charge is very small compared to the quantization of the measurements, such that the quantization itself is statistically significant.

$$\sigma_q^2 = \left(\frac{\partial q}{\partial v_E} \right)^2 \sigma_{v_E}^2 + \left(\frac{\partial q}{\partial v_g} \right)^2 \sigma_{v_g}^2 \dots + \left(\frac{\partial q}{\partial d} \right)^2 \sigma_d^2 + \left(\frac{\partial q}{\partial g} \right)^2 \sigma_g^2 + \left(\frac{\partial q}{\partial V} \right)^2 \sigma_V^2 \quad (14)$$

In total, around 25 videos were obtained, each between one and three minutes in duration such that negatively charged drops were not only easy to find but also easily observable for several cycles of oscillation. The final set of droplets used for analysis produced a total of 44 data points.

Because quantization is difficult to observe for droplets with more than 15 electrons, the initial data points of concern were limited to those with $q < 15e$. Plotting q against drop number, we expect to see a grouping of data points at or near each low integer multiple of the electronic charge. The initial plot is shown in Figure 5(a).

By calculating a weighted average of the separation between points in each grouping, we obtain an estimate for the electronic charge.

$$e_0 = (1.6 \pm 0.2) \times 10^{-19} \text{ C} \quad (15)$$

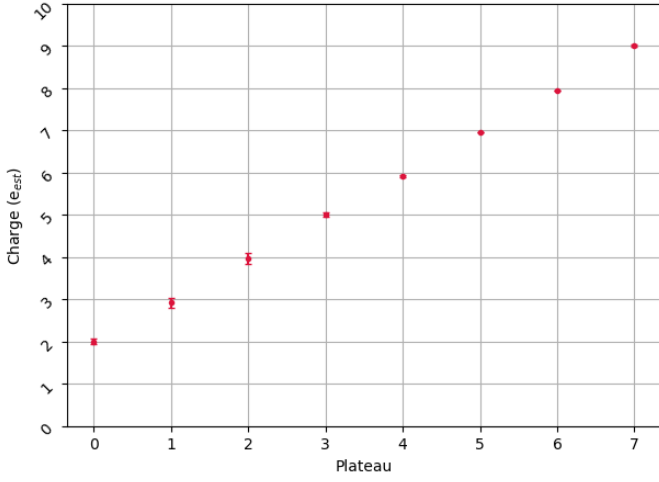
We have obtained the electron charge within $\sim 0.7\%$ of the accepted literature value. However, our uncertainty on the estimate is larger than desirable.

For the sake of attempting to improve our estimate of the charge, we can justifiably remove a few more outliers. The data shown in Figure 5(a) clearly shows a few. If we drop the points with $q > 10e$ (only the droplets with only a few electrons are of much interest anyway) or with $\sigma_q > 2e$, we obtain a cleaner dataset, as shown in Figure 5(b).

The same calculation for the previous estimate now produces a significant improvement.

$$e_1 = (1.58 \pm 0.06) \times 10^{-19} \text{ C} \quad (16)$$

If we plot the weighted average of each “grouping” of these remaining points, we expect to see that the error on



REFERENCES

- [1] R. A. Millikan, “The isolation of an ion, a precision measurement of its charge, and the correction of stokes’s law,” *Phys. Rev. (Series I)*, vol. 32, pp. 349–397, Apr 1911.
- [2] “Millikan Oil Drop Experiment v1.6,” *CarmenCanvas*, 2024.
- [3] B&K Precision, *True RMS Bench Multimeters, 2831E and 5491B*.

Fig. 6: Weighted average and charge uncertainty on each “grouping” of points in the data set. Charge quantization is statistically evident in that no uncertainties overlap between separate groupings.

each weighted average does not overlap with the error of the group above or below. This plot is shown in Figure 6. From this plot, the quantization is quite obvious, and both visually and statistically (most groupings weighted averages are within $\pm 1\sigma$ of some integer multiple of our charge estimate).

V. CONCLUSIONS

The above analysis quite clearly demonstrates the quantization of the charge. If the droplet charge were not quantized, we would expect to measure droplet charges somewhere between these integer multiple plateaus far more often. In fact, there is no overlap between the weighted averages of any two groupings, and most of the groupings have a weighted average which falls within $\pm 1\sigma$ of the integer multiple of our initial guess of the electron charge:

$$e_0 = (1.59 \pm 0.2) \times 10^{-19} \text{ C}$$

This is of course, well within one standard deviation of the accepted value,

$$e = 1.602176634 \times 10^{-19} \text{ C}$$

By combing the data once again to only the highest quality data ($q < 10e$, $\sigma_q < 2e$), we improve our uncertainty, yet we remain within one standard deviation.

$$e_1 = (1.58 \pm 0.06) \times 10^{-19} \text{ C}$$

We have only missed the accepted literature value by approximately 1.4%.

This conclusion in Millikan’s time was quite revolutionary. His ability to measure the electron’s charge with such precision (see Ref. [1]) despite comparatively rudimentary observational tools (he had no camera or tracking software) demonstrates his capabilities and patience as an experimentalist.